DATA 505 (Statistics Using R): Exam 1

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed `.qmd` file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze data about Antarctic penguins from the Palmer Station research program. The data are available in the `palmerpenguins` R package.

```r
# Install and load the required packages
if(!"palmerpenguins" %in% installed.packages()){
  install.packages("palmerpenguins")
}
if(!"ggplot2" %in% installed.packages()){
  install.packages("ggplot2")
}

library(palmerpenguins)
library(ggplot2)

# Load the penguins data
data("penguins")
```

Data Exploration (4 points)

1 (2 points)

Use the `str()` function to examine the structure of the `penguins` dataset.

```
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

How many observations and variables are there?

Answer: There 344 observations with 8 columns and there are total 8 variables.

2 (2 points)

Extract the `bill_length_mm` column from the penguins dataset and store it in a new variable called `bill_lengths`. Then use the `summary()` function to examine this variable.

```
bill_lengths<-penguins$bill_length_mm
summary(bill_lengths)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
  32.10   39.23   44.45   43.92   48.50   59.60       2 
```

Research Question

We want to investigate whether the mean bill length of Adelie penguins differs significantly from 39 mm. This research question is motivated by previous studies suggesting that 39 mm might be a typical bill length for this species.

1 (4 points)

Filter the dataset to include only Adelie penguins and store the result in a new variable called `adelie_penguins`. Remove any rows with missing bill length measurements.

```
adelie_penguins <- penguins[penguins$species == "Adelie" & !is.na(penguins$bill_length_mm), ]
```

2 (3 points)

Extract the bill lengths from the Adelie penguin data and store them in a variable called `adelie_bills`. How many Adelie penguins have complete bill length measurements?

```
adelie_bills<-adelie_penguins$bill_length_mm
length(adelie_bills)
```

```
[1] 151
```

Number of Adelie penguins with complete data: There are 151 Adelie penguins with complete data

3 (5 points)

Calculate the following descriptive statistics for Adelie penguin bill lengths: - Mean - Standard deviation
- Median - 25th and 75th percentiles

```
mean(adelie_penguins$bill_length_mm)
```

```
[1] 38.79139
```

```
median(adelie_penguins$bill_length_mm)
```

```
[1] 38.8
```

```
sd(adelie_penguins$bill_length_mm)
```

```
[1] 2.663405
```

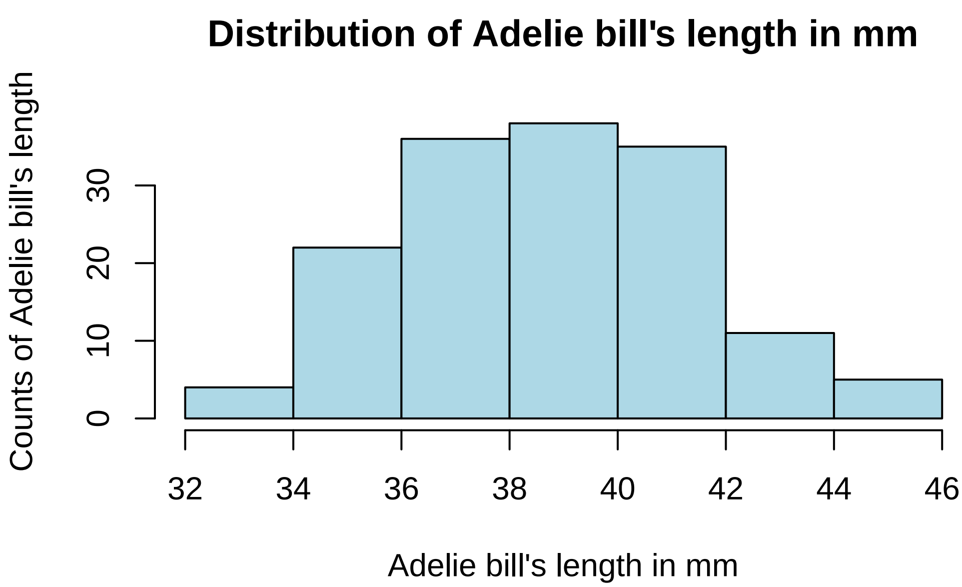
```r
quantile(adelie_penguins$bill_length_mm,props=c(0.25,0.75))
```

```
   0%   25%   50%   75%  100% 
32.10 36.75 38.80 40.75 46.00 
```

4 (5 points)

Create a histogram of Adelie penguin bill lengths using ggplot2 or base R. Include appropriate axis labels and a title.

```r
hist(
  adelie_penguins$bill_length_mm,
  ylab = "Counts of Adelie bill's length",
  xlab = "Adelie bill's length in mm",
  col = "lightblue",
  border = "black",
  main = "Distribution of Adelie bill's length in mm"
)
```



Based on the histogram, briefly describe the distribution of bill lengths (address shape, center, and spread):

Description:

-> Shape : The distribution appears roughly bell-shapes and symmetric,suggesting a normal-like distribution

-> Center : Most bill lengths cluster around 39-40 mm, indicating this is the typical bill length for Adelie

->Speard : The bill length range from approximately 32 mm to 46 mm, showing moderate variability across individuals.

Statistical Analysis

1 (5 points)

Now we will formally test whether the population mean bill length μ differs from 39 inches, i.e. we will test:

$$H_0 : \mu = 39$$

against

$$H_A : \mu \neq 39$$

Perform a two-sided t-test to test these hypotheses. Use $\alpha = 0.05$.

```r
t.test(adelie_penguins$bill_length_mm, mu = 39)
```

```
	One Sample t-test

data:  adelie_penguins$bill_length_mm
t = -0.96246, df = 150, p-value = 0.3374
alternative hypothesis: true mean is not equal to 39
95 percent confidence interval:
 38.36312 39.21966
sample estimates:
mean of x
 38.79139
```

Based on your test results, what is your conclusion? Interpret the results in the context of the research question.

Conclusion: Since the p-value is greater than 0.05, we fail to reject the null hypothesis. This means we do **not** have sufficient evidence to conclude that the population mean bill length of Adelie penguins differs from 39 mm.

2 (4 points)

Define another variable (a numeric vector) `confidence_interval` and assign as its value the two bounds (lower and upper) of a 95% confidence interval for the mean bill length of Adelie penguins. Print the result. Full points are possible only if you do not "hard code" the bounds for the interval, i.e. if you create this vector directly by using R functions applied to the dataset. Half points maximum if you define the vector by typing the entries manually.

```r
# Perform the t-test and extract the confidence interval
test_result <- t.test(adelie_penguins$bill_length_mm, mu = 39)
confidence_interval <- test_result$conf.int

# Print the result
confidence_interval
```

```
[1] 38.36312 39.21966
attr(,"conf.level")
[1] 0.95
```
